UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

Part A: Introduction

1	**Course Title**	Biology for Engineers
2	**Course Code**	BIO3105
3	**Pre-requisites**	Fundamental Biology (Biology in HSC/A Level/Equivalent exams)
4	**Course Type**	Core Course
5	**Credit Hours**	3.00
6	**Contact Hours**	3 Hours/Week
7	**Semester**	6^{th}
8	**Total Marks**	100
9	**Course Instructor & Course Coordinator's Information**	CI: Dr. SM Rafiqul Islam (SMRI) Email: rafiqul@bge.uiu.ac.bd Office: 912 CO: Dr Hasin Anupama Azhari (HAA) Email: ahanupama@ins.uiu.ac.bd Phone Number: +8801711841063 Office: 633(D)
10	**Course Rationale**	This course is required for the students to be able to understand different biological systems and to apply the knowledge to design a biological system within constraints.
11	**Course Objectives**	The objectives of this course are: • To familiarize the basic biological systems and chemistry for life. • To introduce different types of advances in biology. • To familiarize different types of applications in biological field. • To apply the knowledge of biology in real life problems.
12	**Text Book**	1. Biology for Engineers–Arthur T. Johnson (2^{nd} Ed.)
13	**Reference**	1. Applied Cell and Molecular Biology for Engineers–Editors: Gabi Nindl Waite and Lee R. Waite 2. Biology for Dummies–Rene Fester Kratz and Donna Rae Siegfried (2^{nd} Ed.) 3. Biology for Engineers–G. K. Suraishkumar 4. A New Biology for the 21^{st} Century by The National Research Council, The National Academic Press, Washington D.C. 5. Molecular Biology of the Cell–Bruce Alberts (7^{th} Edition) 6. Systems Biology: A Textbook (2^{nd} Edition) by Edda Klipp

Part B: Content of the Course

14	**Course Contents (approved by UGC)**	**1. <u>Introduction:</u>** The Basics of Life: Importance of biology for engineers. **2. <u>Cell Structure and cell division:</u>** Understanding cell structure, cell function, mitosis, meiosis, cell transport mechanism, fluid mosaic model **3. <u>Genetics and Genomics:</u>** DNA, RNA structure and replication, Central Dogma, Mendel's Laws of Inheritance, Chromosomes and genes, Gene expression

(transcription, translation), Genetic variation and mutations, Genetic Disorders: (Karyotypes, Chromosomal disorders, Polyploidy, Trisomy-down syndrome)

4. **Molecular Biology and Biotechnology:** Biotechnological application-PCR, Gel electrophoresis, RDNA CRISPR and gene editing, Personalized medicine and pharmacogenomics, Genome sequencing (Sanger, NGS), Human Genome project, Genomic data ethics and privacy

5. **Immunity:** The immune system and microbial infection, Innate and Adaptive immunity, Components of adaptive and innate immune system, Vaccination

6. **Structural and Systems Biology:** organic molecules and chirality, biological macromolecules and structure hierarchy (of proteins, carbohydrates, lipids, and nucleic acids), Structure-function relationship (how 3D shape defines biological function), applications (drug design, protein engineering), Biological networks (metabolic, gene regulatory, signaling, neural network), omics (genomics, epigenomics, proteomics, metabolomics), application of synthetic biology

7. **Biological Data/ database:** Introduction to biological databases, Classification and functions of biological databases, Tools used for bioinformatics studies, Database searching: GenBank, Blast, PDB, UniProt, Primer designing, Protein Structure Prediction, ML based biological databases, Tools related to whole genome sequence analysis: Galaxy server

8. **Brain Computer Interface:** Introduction to Neurophysiology and Neuroanatomy (briefly cover the nervous systems, neurons, and action potential), Neurotransmitters vs Neuropeptides vs neurohormones, Glial cells and their functions, Insights into neuroengineering (Brain-Computer Interface–BCI, Artificial Neural Network – ANN)

15 Course Outcomes (COs)

COs	Description
CO1	Understand the basics and importance of biology in daily life.
CO2	Describe the parts of a cell and explain how cells divide and grow.
CO3	Explain how DNA and RNA carry genetic information in living things, how traits are inherited and anomalies cause diseases.
CO4	Understand how molecular biology and biotechnology are used to improve health, agriculture, and the environment.
CO5	Describe how the immune system protects the body from diseases.
CO6	Understand how biological molecules work and how synthetic biological systems can be designed
CO7	Use biological databases to find and understand scientific information and study the genome, proteome, etc.
CO8	Describe how the brain functions as the most efficient processor known to man

16 Mapping of COs and Program outcomes

COs	Program Outcomes (POs)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓					✓	✓					✓

CO2	✓	✓		✓								
CO3	✓	✓				✓						
CO4			✓		✓	✓	✓					
CO5	✓	✓				✓						
CO6	✓		✓	✓	✓							
CO7		✓		✓	✓							✓
CO8	✓	✓			✓				✓			✓

17 Mapping COs with Teaching-Learning and Assessment Strategy

Class	Topics/Assignments	Course Outcomes (COs)	Reading Reference	Teaching-Learning Strategies	Assessment Strategies
1, 2	Introduction: Why Biology in Engineering; Play with Biology; Scope for Engineers	1	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
3, 4	Cell Structure and Function: Plant and animal cells, fluid mosaic model, protein factories of the cell, nucleated and enucleated cells	2	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
5,6	Cell Division, Cell cycle	2	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
7,8	Genetics and Genomics, Genetic Disorders	3	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
9,10,11	Applications of Molecular Biology and Biotechnology	4	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
12	Review				
13,14	The Body's Control Mechanisms and Immunity, Homeostasis, Immunoglobulin	5	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
15, 16, 17	System and Structural biology: Macromolecules, Structure-function relationship, Biological networks,	6	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam

	Omics, Synthetic Biology				
18, 19, 20	Biological Data/ database	7	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
21, 22, 23	Brain-Computer Interface–BCI: Neurotransmitters vs Neuropeptides, Glial cells and their functions. Insights into neuroengineering	8	Lecture slides	Lecture/ Exercise	Class Tests/Assignments/Quizzes/Exam
24	Review				

Part C: Assessment and Evaluation Methods

Assessment Strategy	Assessment Types	Marks
Formative Assessment	Attendance	5%
	Assignments	5%
	Class Tests	20%
Summative Assessment	Mid Term	30%
	Final Exam	40%

Grading System

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	An ability to apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.

PO3	An ability to design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO4	An ability to conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO5	An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations
PO6	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO8	An ability to apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO9	An ability to function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO10	An ability to communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one’s work as a team member or a leader to manage projects in multidisciplinary environments.
PO12	An ability to recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.